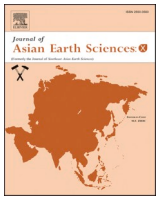
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Assessing the performance of machine learning and analytical hierarchy process (AHP) models for rainwater harvesting potential zone identification in hilly region, Bangladesh

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ABSTRACT

Water scarcity in hilly regions presents unique challenges, particularly in Bangladesh, where obtaining fresh drinking water has become difficult to access. This study aims to evaluate the potential zones for rainwater harvesting (RWH) using machine learning (ML) algorithms and geospatial analysis. Specifically, four ML algorithms—random forest (RF), boosted regression trees (BRT), k-nearest neighbors (KNN), and naïve bayes (NB)—alongside the analytical hierarchy process (AHP) were employed to delineate potential RWH zones in the Chattogram hilly districts, including Chattogram, Rangamati, Bandarban, Khagrachari, and Cox's Bazar. Eleven influencing factors were considered: aspect, distance from road, drainage density, elevation, hill shade, lineament density, land use/land cover (LULC), slope, topographic wetness index (TWI), rainfall, and geology. Inventory data from the study area, consisting of 135 suitable and 135 non-suitable points, were randomly split, with 70% used for training the models and the remaining 30% for validation using the area under the curve (AUC) values. The southern regions are highly suitable for harvesting rainwater. Among the five models, BRT and RF demonstrated superior performance with AUC values of 0.93 for both models. In contrast, the AHP method yielded the lowest AUC value at 0.82. Notably, drainage density and elevation emerged as the most influential factors in constructing these models. The application of machine learning algorithms has enhanced the precision of rainwater harvesting zone estimate systems by examining diverse aspects. The findings of this study can provide valuable insights for policymakers in making informed decisions regarding RWH in these regions.

1. Introduction

Water scarcity is a critical issue affecting many regions globally, especially in developing nations where the increasing population and economic growth are driving up water demand. Presently, around 3.6 billion people, which accounts for over half of the global population, live in areas experiencing water shortages. This number is expected to increase to between 4.8 and 5.7 billion by 2050 (UN-Water, 2018). However, only about 2.5 % of the world's total water volume is freshwater, with less than 1 % being usable freshwater, highlighting the finite nature of this vital resource (Blanco and Lal, 2023). About one-third of the world's freshwater comes from underground sources, with residential, agricultural, and industrial uses accounting for 36 %, 42 %, and 27 % of groundwater usage, respectively (Kumar et al., 2016; Taylor et al., 2013).

Rain is a primary source of groundwater, replenishing it as it seeps through soil and rock. The leading global pressures on clean water availability are climate change, population growth, rapid industrialization, and excessive groundwater consumption. Scientists worldwide have explored various alternative strategies to address water scarcity. In

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many water-stressed regions, rainwater harvesting (RWH) has become a prevalent practice, recognized as one of the most cost-effective and environmentally beneficial water conservation methods (Al-Abadi et al., 2017; Preeti et al., 2022; Rana and Moniruzzaman, 2023). The RWH structures help to reduce runoff rates while enhancing groundwater efficiency and aquifer recharge (Kadam et al., 2012). The main challenges in implementing RWH techniques are associated with factors such as rainfall patterns, intensity and distribution, runoff characteristics, soil water storage capacity, reservoir capacity, agroforestry systems, available technology, and socio-economic conditions (Alkaradaghi et al., 2022).

Although, Bangladesh is a riverine country with over three hundred rivers, the supply of fresh water is insufficient to meet all human needs due to geopolitical challenges, construction projects, climate issues, and other factors. Environmental and hydrological conditions suggest that the Chattogram region of Bangladesh is well-suited for RWH. Implementing RWH in this area can significantly enhance water availability for various uses (i.e., drinking, agriculture), contributing to the region's goal of achieving self-sufficiency. Chattogram is one of the most critical regions for effective management of water resources, sustainable development, and environmental conservation. The RWH has emerged as a viable solution to address water scarcity by capturing and storing rainwater, reducing the dependence on traditional water sources.

Recent advancements in geographic information systems (GIS) and remote sensing (RS) (Hashim and Sayl, 2021), and multi-criteria decision-making (MCDM) (Mohammed et al., 2023) techniques have provided new opportunities for identifying suitable locations for RWH structures. These technologies enable the integration and analysis of various environmental and socio-economic factors to create comprehensive suitability maps for RWH (Sayl et al., 2016; Singh et al., 2017; Steinel et al., 2016). This approach ensures that the selected sites for RWH are not only hydrologically viable but also socio-economically beneficial (Abdellah, 1979; Assaf et al., 2021; Seidler and Slotkin, 1976).

Research around the globe has leveraged geospatial techniques to pinpoint optimal sites for RWH structures. For instance, in Iran, Karimi and Zeinivand (2021) used GIS and the analytical hierarchy process (AHP) to combine runoff, land use, and other factors to identify potential RWH zones in the Kakareza watershed. Toosi et al. (2020) incorporated socio-economic elements such as proximity to infrastructure and natural features for RWH site selection in the Mashad Plain Basin in Saudi Arabia. Similarly, studies in the USA (Sekar and Randhir, 2007), western India (Kadam et al., 2012), and Sydney (Preeti et al., 2022) have shown the effectiveness of using multiple spatial layers in GIS to determine the best locations for RWH. Moreover, Akter and Ahmed (2015) highlighted the interaction of local knowledge and geoinformatics in urban Bangladesh while Rana and Moniruzzaman (2023) analyzed multiple spatial parameters to assess RWH suitability in northwestern Bangladesh.

The machine learning (ML) techniques are increasingly being incorporated into RWH systems to boost their efficiency and predictive capabilities. Research suggests that ML models can enhance the design and management of RWH systems by analyzing past rainfall data and forecasting future patterns, thereby improving water collection strategies (Halder and Bose, 2024). For example, sophisticated algorithms have been utilized to evaluate the potential yield of rainwater, taking into account various environmental factors, which supports more informed decision-making (Rao et al., 2024). Furthermore, studies have shown that ML can enable real-time monitoring and management of RWH, facilitating adaptive responses to changing weather conditions (Gaurav et al., 2021). Despite the extensive research on the application of GIS, RS, and MCDM in water resource management, there is a notable gap in the literature regarding the specific application of these techniques in the context of Chattogram, Bangladesh. Most existing studies have focused on regions with different climatic and geographical characteristics, such as arid regions or areas with distinct hydrological patterns. Moreover, while some research has been conducted in urban areas of Bangladesh, there is limited information on the rural and *peri*-urban contexts, particularly in regions like Chattogram that are prone to both water scarcity and excess. The novelty of this research lies explicitly within this research gap.

The most recent research regarding RWH explores the location of RWH infrastructure (Chen et al., 2020; Du et al., 2024; Ullah et al., 2024). The RWH potentiality was investigated in saline regions (Abdullah et al., 2024; Saha et al., 2024), and rooftops (Hossan et al., 2023; Nipun et al., 2024). The ML approach used for potential RWH in the hilly region was still unexplored in Bangladesh. The primary objective of this study is to develop suitability maps for RWH structures in Chattogram, Bangladesh, using ML and MCDM approaches. These two approaches weigh the strengths of each tool to provide a robust framework for identifying optimal sites for harvesting rainwater. By incorporating eleven thematic layers, such as land use, soil type, slope, rainfall patterns, and socio-economic factors, the study aims to prioritize locations that maximize the benefits of RWH while minimizing potential environmental impacts. The outcome of this research will help policymakers and planners use the appropriate ML algorithm to identify the RWH zone.

2. Material and methods

2.1. Study area

The study area, located in southeastern Bangladesh between latitudes 20°45′ N to 23°45′ N and longitudes 91°25′ E to 92°40′ E, encompasses the chattogram hill tracts (CHT), Chattogram, and Cox's Bazar districts in Bangladesh (Fig. 1). About 12.3 million people live within this area, most of whom live in the Chattogram district. The CHT covering Khagrachari, Rangamati, and Bandarban, feature elevations from 300 to over 1000 m with dense forests. Most of the population of this area are small tribes and indigenous communities. The region receives the highest amount of rainfall among all other areas of Bangladesh, most of which occurs during the rainy season (June to September). Cox's Bazar receives the highest annual rainfall in the region, around 3500 mm, while the CHT and Chattogram districts also receive substantial rainfall, exceeding 2500 mm annually. The average temperature of this region ranges between 13 °C to 26 °C in winter and 25 °C to 36 °C in summer seasons (Islam et al., 2020). Most of the landforms of this region are undulating hilly areas. The predominantly hilly landscape of the area limits access to groundwater and other available water resources.

2.2. Inventory data

The researchers utilized inventory datasets to delineate RWH zones. A comprehensive field investigation gathered 135 locations categorized as suitable (class value 1) and 135 locations categorized as non-suitable (class value 0) for RWH mapping. These points were randomly split, with 70 % assigned to a training set and the remaining 30 % to a testing set. This testing set enabled the validation and accuracy assessment of the models employed in the study.

2.3. Influencing factors of rainwater harvesting

This study integrates 11 influencing factors to evaluate the potential for RWH in the Chattogram hill districts, namely Chattogram, Khagrachari, Bandarban, Rangamati, and Cox's Bazar. These factors include aspect, distance from the road, drainage density, elevation, hill shade, lineament density, land use/land cover (LULC), slope, topographic wetness index (TWI), rainfall, and geology (Table 1). The study's overall methodology is illustrated inFig.2 . The details of the influencing factors are given inFig. 3 ..

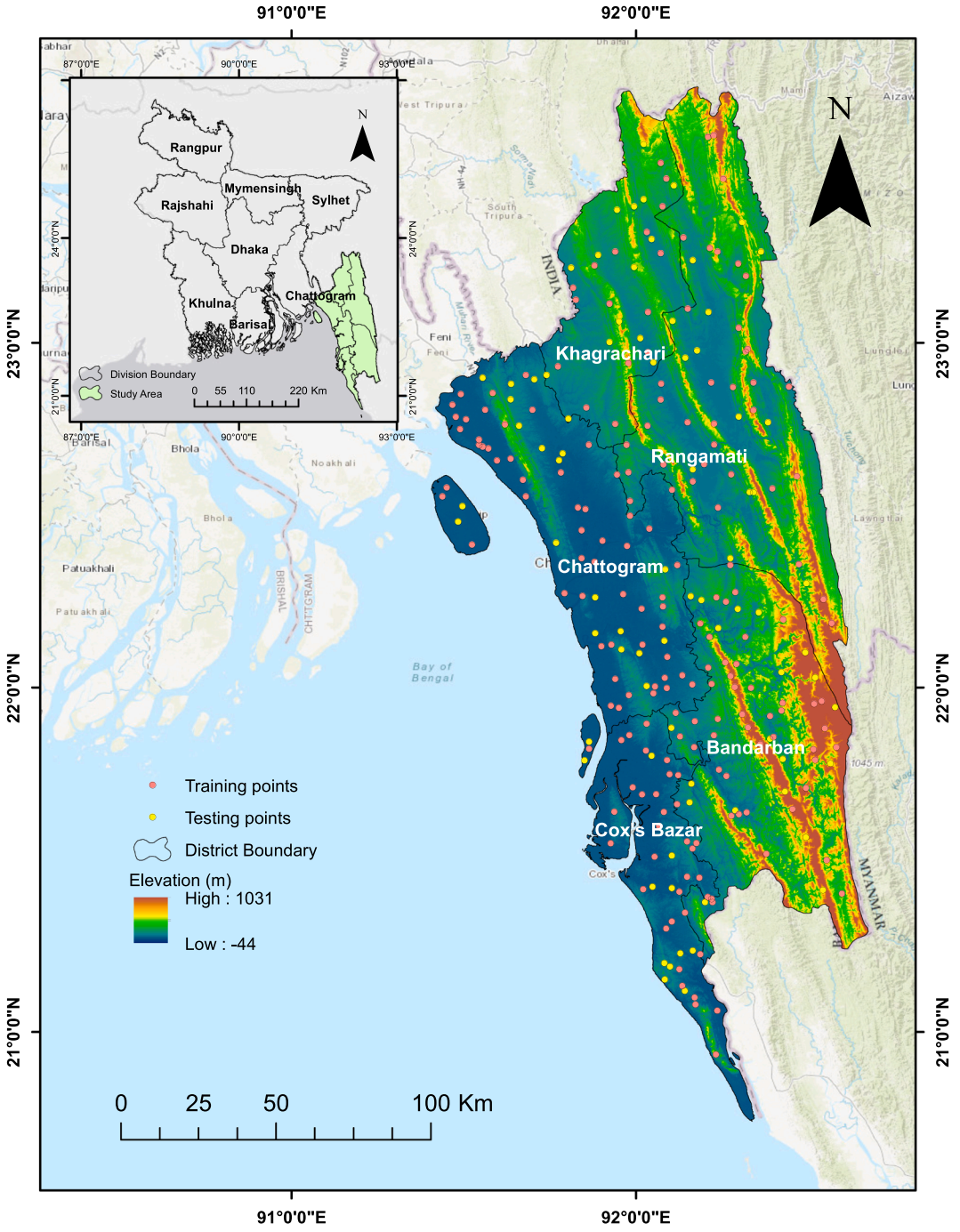


Fig. 1. A map delineating the research region in southeastern Bangladesh, which includes the Chattogram Hill Tracts (CHT) and the districts of Chattogram and Cox's Bazar. The inset map illustrates the geographical position of the research region inside Bangladesh. The allocation of training (yellow) and testing (red) points used in the rainwater harvesting (RWH) suitability modelling is also shown. The colour gradient represents elevation data in meters.

2.3.1. Elevation

Elevation significantly impacts rainfall distribution and runoff direction, thereby influencing the potential for rainwater collection. In the study area, elevation exhibits considerable variation, ranging from −44 m (indicating river and lake levels) to 1031 m (representing the highest points in the CHT) (Fig. 3a). Higher elevations tend to receive increased rainfall. In contrast, lower-lying areas provide flattened surfaces conducive to the placement of storage tanks.

2.3.2. Topographic wetness index (TWI)

The TWI is a measure of the potential for water accumulation on a topographic surface. Higher TWI values (2144–26435) suggest areas with lower slopes and a higher likelihood of water infiltration and saturation (Fig. 3b). These areas might be suitable for locating RWH structures as they have the potential to capture and store more rainwater.

2.3.3. Slope

The slope indicates the gradient of the terrain. The study area displays notable variability, ranging from flat surfaces (0 degrees) to very steep slopes (up to 73.8 degrees) (Fig. 3c). Steeper slopes accelerate water runoff, potentially decreasing the availability of rainwater for collection. In contrast, gentler slopes (lower degrees) provide optimal conditions for RWH tanks, promoting water infiltration and storage.

Table 1
Details of the influencing factors utilized in the study.

Data	Sources	Resolution
Digital Elevation Model (DEM)	(USGS 1 arc second), UTM-45, WGS 1984. (https://earthexplorer.usgs.gov/)	30 *m*
Land use land cover map (LULC)	Landsat 8 OLI TIRS (https://earthexplorer.usgs.gov/)	30 *m*
Hillshade	DEM (USGS 1 arc second), UTM-45, WGS 1984. (https://earthexplorer.usgs.gov/)	30 *m*
Slope	DEM (USGS 1 arc second), UTM-45, WGS 1984. (https://earthexplorer.usgs.gov/)	30 *m*
Drainage density	DEM (USGS 1 arc second), UTM-45, WGS 1984. (https://earthexplorer.usgs.gov/)	30 *m*
TWI	DEM (USGS 1 arc second), UTM-45, WGS 1984. (https://earthexplorer.usgs.gov/)	30 *m*
Soil types	FAO/UNESCO Soil Map of the World (https://data.apps.fao.org/)	30 *m*
Rainfall	Bangladesh Meteorological Department (BMD) (https://bmd.gov.bd/)	30 *m*
Aspect	DEM (USGS 1 arc second), UTM-45, WGS 1984. (https://earthexplorer.usgs.gov/)	30 *m*
Distance from Road	Euclidean distance Using ArcGIS 10.8	30 *m*

2.3.4. Aspect

Aspect denotes the compass direction that a slope faces, ranging from −1 to 359.845 degrees (Fig. 3d), indicating slopes oriented in all directions. Aspects influence factors such as sun exposure and evaporation rates, which in turn impact the quality and quantity of harvested rainwater. South-facing slopes may experience more excellent evaporation rates, whereas north-facing slopes might provide cooler conditions and potentially higher-quality stored water.

2.3.5. Drainage density

Drainage density reflects the proximity of streams and channels within a given area. Values range from 0 (indicating no defined drainage channels) to 1.054 (suggesting a landscape with dense, closely spaced stream networks). Higher drainage density typically results in faster rainwater runoff, potentially decreasing the amount available for harvesting. Conversely, lower drainage density may signify areas where water accumulates, potentially making them suitable for rainwater collection (Fig. 3e).

2.3.6. Distance from road

Distance from the road measures the proximity of potential RWH sites to road networks. The range spans from 0 to 48.20 km, encompassing locations ranging from immediately adjacent to roads to more remote areas (Fig. 3f). Proximity to roads impacts factors such as transportation costs associated with the installation and upkeep of RWH systems. Sites located closer to roads are generally more accessible and may entail fewer logistical challenges.

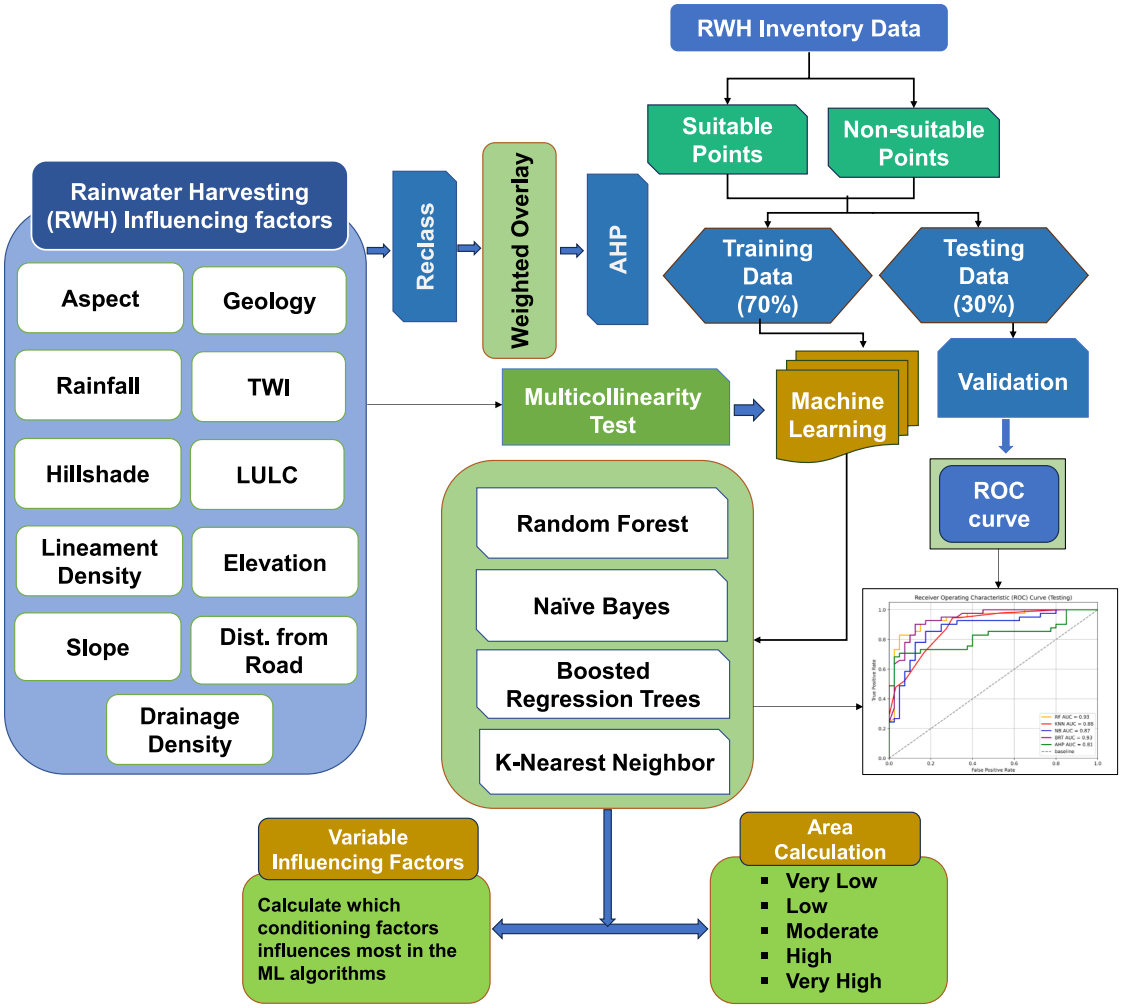


Fig. 2. Rainwater harvesting (RWH) potential in the research region is represented spatially by eleven different parameters that influence it. Each panel depicts; (a) Elevation (m), (b) Topographic Wetness Index (TWI), (c) Slope (°), (d) Aspect (°), (e) Drainage Density, (f) Distance from Road (km), (g) Rainfall (mm), (h) Soil Type, (i) Hillshade (°), (j) Lineament Density, (k) Land Use/Land Cover (LULC).

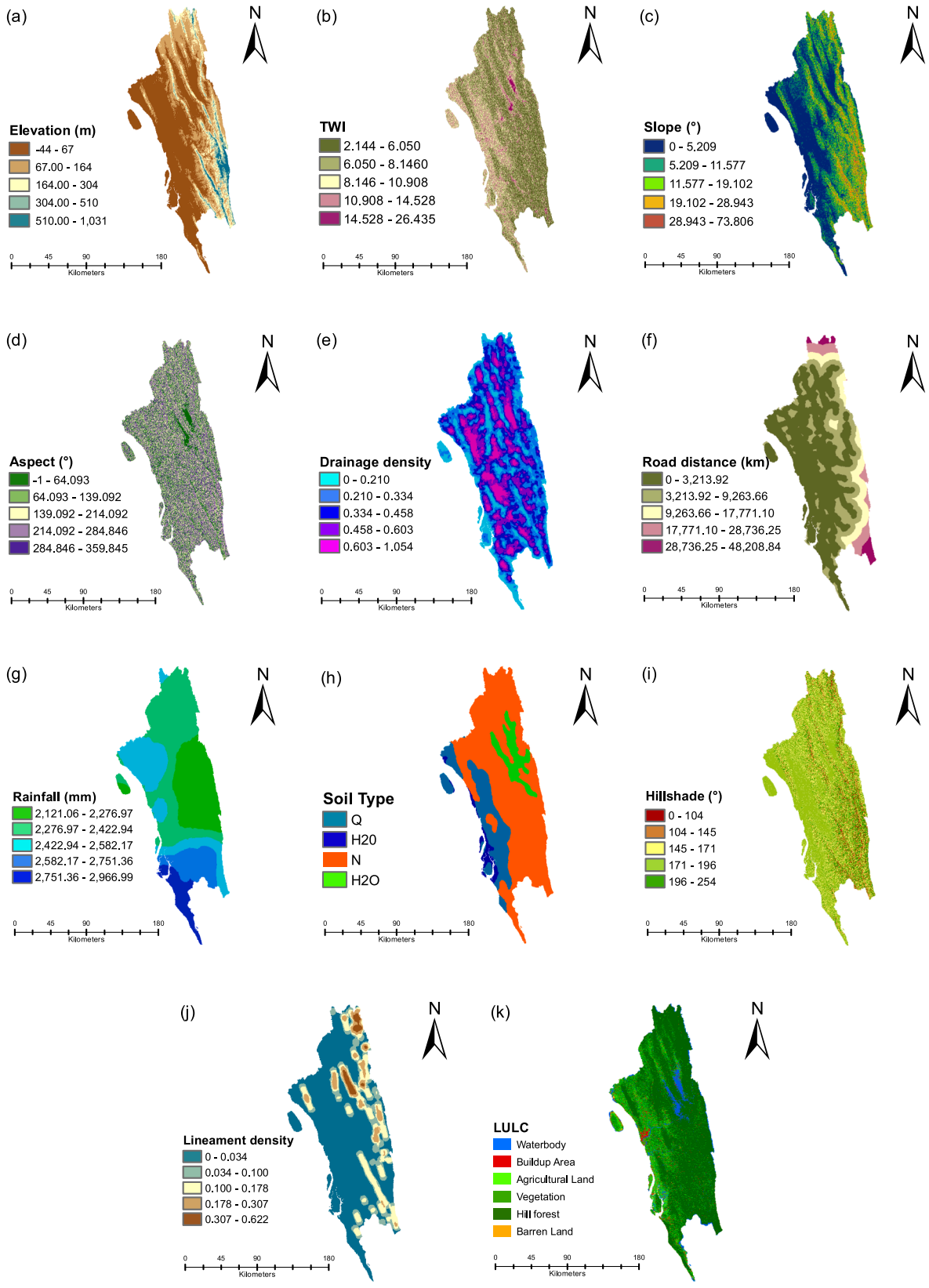


Fig. 3. Flowchart detailing the methodology employed in this study to identify appropriate zones for rainwater harvesting (RWH) in the Chattogram Hill Tracts (CHT) region of Bangladesh. The procedure encompasses gathering data, conducting pre-processing (which includes reclassification and weighted overlay through AHP), training and validating machine learning models, and ultimately producing the RWH suitability map.

2.3.7. Rainfall

Rainfall refers to the quantity of precipitation received in a particular location. The range within the study area (2121 to 2966 mm) signifies varying annual rainfall totals (Fig. 3g), indicating areas with potentially lower or higher amounts of precipitation. Rainfall is a pivotal factor influencing the potential for RWH. This region was experiencing higher rainfall levels, which offered increased resources for collection and storage.

2.3.8. Soil

Soil properties affect infiltration, drainage, and water storage capacity (Fig. 3h). Data from the food and agricultural organization (FAO) categorizes the study area into different soil types, each with distinct characteristics. Sandy soil typically exhibits higher infiltration rates but lower water-holding capacity, whereas clay soils tend to have lower infiltration rates but higher water retention potential. Understanding these soil types is crucial for assessing the suitability of locations for RWH and predicting potential challenges related to infiltration.

2.3.9. Hill shade

Hill shade indicates the relative amount of shading received by a location from surrounding topography. The range from 0 to 254 degrees suggests varying degrees of shading throughout the day across the study area. Hill shade influences factors such as evaporation rates; areas with more excellent shade may experience reduced evaporation, potentially enhancing the quality of stored rainwater (Fig. 3i).

2.3.10. Lineament density

Lineament density denotes the frequency of linear geological features, such as faults or fractures, within the bedrock. The range from 0 to 0.622 indicates variability across the study area (Fig. 3j), from regions lacking detected lineaments to those with higher densities. Lineaments play a significant role in influencing groundwater flow patterns and infiltration rates, and understanding lineament density aids in assessing potential subsurface water storage capabilities and their interaction with collected rainwater.

2.3.11. Land use/Land cover (LULC)

This dataset categorizes the surface characteristics of the land into likely six classes: waterbody, built-up area, agricultural, vegetation, hill forest, and barren land. The LULC classification is vital for assessing the suitability of the various regions for RWH. Collecting rainwater from impervious surfaces such as rooftops in built-up areas may be more efficient than in vegetated areas. However, water quality from different LULC types must be considered (Fig. 3k).

2.4. Analytical hierarchy process (AHP)

Saaty (1977), invented the AHP, a method used to calculate the importance of different map layers and to combine their properties. The AHP is an invaluable technique for making decisions and helps to ensure the coherence of attributes put forth by decision-makers. The process includes the pairwise comparison of the factors to identify the hierarchy of their influence (Saaty, 1977). The AHP is frequently employed in the study of MCDM due to its standard accuracy, straightforwardness, ease of use, and clarity (Saaty, 2005). An essential component of this approach entails the identification of elements that have a substantial impact on RWH and organizing them into appropriate subcategories. The parameters discovered in the pairwise comparison matrix were evaluated using the significance scale proposed by Saaty (1980).They were assigned ratings on a scale ranging from 1 to 9. The AHP rating system assigns weights on a continuous 9-point scale: (1/9, 1/8, 1/7, 1/6, 1/5, 1/4, 1/3, 1/2, 1, 2, 3, 4, 5, 6, 7, 8, 9). In this scale, factor weight values greater than 1 indicate higher significance, values less than 1 signify lower significance, and a value of 1 denotes equal importance when compared to another factor (Ahmed, 2015). The potential for RWH was incorporated into these assigned weights and their pairwise comparison matrices, constructed using the AHP as developed by Saaty (1980). Subsequently, these weights were normalized using the eigenvector technique, and their consistency was evaluated by calculating the consistency ratio (Eq. (1).

$$CI = \frac{\mu max - n}{(n-1)} \quad (1)$$

Where μmax is the most significant or principal eigenvalue of the matrix, and n is the order of the matrix. It is stated that CR ratings greater than 0.10 should be re-evaluated (Saaty, 1977) (Eq. (2).

$$CR = \frac{CI}{RI} \quad (2)$$

Where RI is the average of the resulting CI depending on the order of the matrix given by Saaty (1977).

To assign weight values, a thorough assessment was performed, integrating insights from existing literature, on-site field observations, and expert opinion. After determining the weight values, all parameters were reclassified according to the influence of different classes of each parameter on RWH, and grid maps in raster format with a 30 × 30 m resolution were generated. In the final stage, the weighted overlay method was applied, utilizing the assigned weight values for all parameters to produce an RWH suitability map.

2.5. Machine learning (ML) algorithms

This study focused on four supervised ML algorithms—random forest (RF), naïve bayes (NB), boosted regression trees (BRT), and k-nearest neighbor (KNN)—to determine the RWH probability zones in the hilly region of Chattogram.

2.5.1. Random forest (RF)

RF is an ML ensemble method utilized for both classification and regression tasks (Malakar et al., 2021). This technique consists of multiple decision trees, which together mitigate the overfitting problem inherent to individual decision trees. During the training phase, several unique bootstrap samples are generated from the original dataset. Approximately one-third of the data is reserved as test cases, while the remaining data is used for training. The out-of-bag error, an unbiased estimate of the test error, is then computed based on the reserved test cases, thereby indicating the RF model's predictive performance (Breiman, 2001). To achieve stable and accurate results, the outputs of the individual decision trees are aggregated using the majority vote technique (Singh et al., 2021).

2.5.2. Naïve bayes (NB)

The NB is a probabilistic method grounded in Bayesian theory that makes decisions through straightforward probability computations, assuming that all variables are conditionally independent of one another. This assumption simplifies the representation and estimation of conditional probabilities, enhancing the effectiveness of NB (Ouyang et al., 2021; Zhao and Li, 2020). By presuming conditional independence, NB reduces the number of variables that need estimation, thereby facilitating the computation of conditional probabilities using a Bayesian classification approach. This assumption arises from the need to assess probabilities associated with the training data despite the presence of many attribute/value combinations that may be absent or underrepresented (Chen et al., 2020; Zhang et al., 2020). The NB effectively addresses this issue through its conditional independence assumption. The classifier is capable of handling both binary and multiclass categorization problems. Its simplicity contributes to its success in various real-world applications despite its basic assumption of feature independence. To apply NB in RWH potential modeling and mapping, X = (x_1, x_2…x_n) represents a vector of independent explanatory factors. The Eq. (3) estimated the probability of RWH zone (P (Cj|x1…, xn)):

$$P(C_j|X) = \frac{P(C_j)P(X|C_j)}{P(X)} \quad (3)$$

2.5.3. Boosted regression trees (BRT)

The BRT model is constructed by integrating multiple simple traditional regression tree models, functioning as a decision tree tool. By leveraging the advantages of boosting techniques alongside conventional regression trees, BRT enhances predictive accuracy. The BRT model offers several benefits, including the ability to fit complex nonparametric relationships, manage various types of predictor variables, and automatically handle interaction effects among variables (Pourghasemi et al., 2020). Moreover, this model mitigates overfitting and allows for the management of missing data without the need for prior data modification.

2.5.4. K-Nearest neighbor (KNN)

The KNN algorithm classifies a sample by considering the majority class among its k-nearest data points, such as determining if a sample corresponds to an RWH zone. To evaluate sample similarity, a distance metric indicating the similarity or dissimilarity of conditioning factors is used; typically, the value of "k" is a small positive integer (Sameen et al., 2020). When conditioning factors are similar, the feature distance is small. Despite its simplicity regarding hyperparameters, the computational demands of the KNN model increase significantly with a large sample size. To assess probability based on a defined set of conditioning factors, the algorithm calculates the distance to each training data point within a multidimensional space. It then identifies the k nearest data points, which are used to determine the RWH zone. The majority class among these neighbors decides the classification of the new data point (Cheng et al., 2013).

2.6. Validation

In the development of an RWH potential zone map, the evaluation of predictive accuracy is a crucial step. The area under the receiver operating characteristic (ROC), area under curve (AUC) is a highly effective method for assessing the reliability and quality of susceptibility maps. This widely adopted approach measures the performance of predictive models, providing a robust indication of their predictive capabilities (Chuvieco et al., 2010; Yesilnacar and Topal, 2005). The interpretation of AUC values is as follows: an AUC equal to or below 0.5 indicates no improvement in prediction accuracy over random chance. An AUC in the range of 0.7 to 0.9 signifies a reasonably good agreement between the model and the observed data. AUC values equal to or greater than 0.9 suggest an optimal scenario where the model's predictions closely align with the actual potential zone of RWH.

3. Results

3.1. Correlation matrix and multicollinearity test

Liu et al. (2018) noted that correlations above 0.7 typically indicate collinearity concerns. In this study, correlations between factors influencing RWH potential showed significant relationships: slope and elevation had a strong positive correlation (r = 0.69), while elevation displayed moderate positive correlations with geology (r = 0.35), lineament density (r = 0.33), LULC (r = 0.32), and distance from roads (r = 0.60) (Fig. 4). The TWI was negatively correlated with slope (r = -0.53) and elevation (r = -0.34), indicating that flatter, lower-elevation areas retain more moisture. Drainage density correlated moderately with TWI (r = 0.27) and negatively with slope (r = -0.30), suggesting flatter, wetter areas have denser drainage networks. Lineament density showed positive correlations with slope (r = 0.40) and distance from roads (r = 0.50), suggesting rugged areas with more geological lineaments are further from roads. To address potential multicollinearity, variable inflation factor (VIF) and tolerance tests were conducted, with all VIF values below 10, indicating no significant multicollinearity (Table 2). The slope and elevation had the highest VIF values (2.65 and 2.34) but remained within acceptable limits, while tolerance values confirmed no multicollinearity issues, with slope and elevation having the lowest

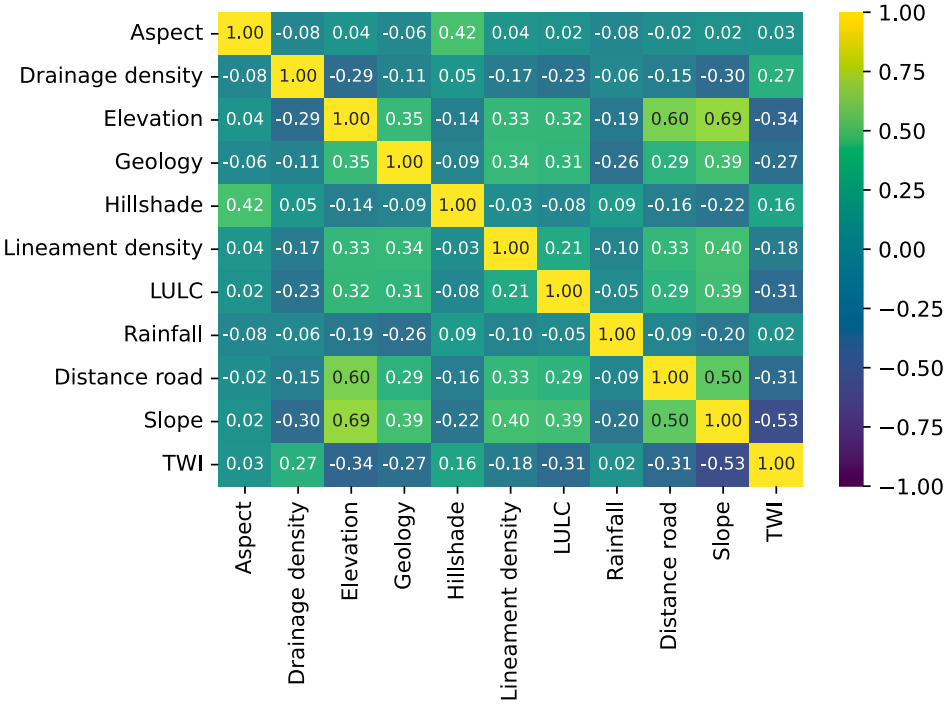


Fig. 4. The correlation matrix showing the magnitude of the relationship among eleven parameters (Aspect, Drainage density, Elevation, Geology, Hillshade, Linement density, LULC, Rainfall, Distance from road, Slope, and TWI) used for RWH. The values of correlation range from + 1 to −1, as shown in the right side of the main original figure. The highest positive correlation was found between slope and elevation, whereas the highest negative correlation was found between slope and TWI.

Table 2
Multicollinearity (MC) test of the Rainwater harvesting conditioning factors.

Parameters	Tolerance	VIF
Aspect	0.782	1.279
Drainage Density	0.837	1.195
Elevation	0.427	2.343
Geology	0.726	1.378
Hillshade	0.752	1.329
Lineament Density	0.775	1.291
LULC	0.786	1.272
Rainfall	0.866	1.154
Distance from Road	0.597	1.675
Slope	0.376	2.659
TWI	0.676	1.480

values (0.37 and 0.42). Other parameters had VIF values between 1.15 and 1.67 and tolerance values between 0.59 and 0.86, indicating no significant multicollinearity. These results suggest that all predictors can be reliably used in the classification model to assess RWH potential without significant risk of multicollinearity affecting the model's stability.

3.2. Rainwater harvesting zone

The study of RWH zones in the Chattogram hilly region utilized five different models—RF, KNN, NB, BRT, and AHP—to analyze the variations in predictions influenced by geographical direction and area. The RWH was divided into five classes: very low, low, moderate, high, and very high (Fig. 5) based on natural breaks (Jenks) on ArcGIS 10.8.

The BRT model indicates "very low" suitability in the northern region, encompassing 31.03 % (6091.45 km^2), "low" suitability in the central region at 20.87 % (4096.69 km^2), and significant "very high" suitability in the southern region at 25.62 % (5028.47 km^2). The KNN model identifies the northern region as "very low" at 10.45 % (2051.84 km^2), with the central region predominantly "low" at 29.38 % (5767.85 km^2) and the southern region as "very high" at 22.90 % (4496.27 km^2). The NB model shows a mixed distribution with "very low" at 18.64 % (3659.31 km^2) and "low" zones at 20.17 % (3958.80 km^2) in the north and central regions, while "moderate" at 22.65 % (4446.36 km^2) and "very high" at 21.72 % (4264.71 km^2) zones are more evenly spread across the regions.

The RF model predicts the highest "low" suitability in the central region at 30.20 % (5927.97 km^2), "very low" suitability in the north at 18.89 % (3708.85 km^2), and a combination of "moderate" at 16.87 % (3311.90 km^2) and "very high" at 17.99 % (3531.35 km^2) in the southern region. The AHP model predominantly forecasts "moderate" suitability across the entire region at 61.99 % (12169.54 km^2), with "high" suitability zones in the central and southern areas at 23.82 % (4675.81 km^2) and negligible "very low" zones at 0.14 % (27.69 km^2) (Table 3).

These models suggest that the northern region is generally less suitable for harvesting rainwater (Fig. 6), the central region ranges from low to moderate suitability, and the southern region exhibits higher potential. This highlights the importance of targeted resource management based on geographical direction.

3.3. Validation of the models

The classification performance of the five models can be observed based on their AUC values. The BRT and RF models both achieved the highest AUC of 0.93, indicating high accuracy and reliability in differentiating between categories. The KNN model follows with a respectable AUC of 0.88, while the NB model has a slightly lower AUC of 0.87 (Fig. 7). Both KNN and NB demonstrate adequate performance, though with less precision compared to BRT and RF. The AHP model has the lowest AUC of 0.82 (Table 4), reflecting moderate performance and making it the least effective among the models.

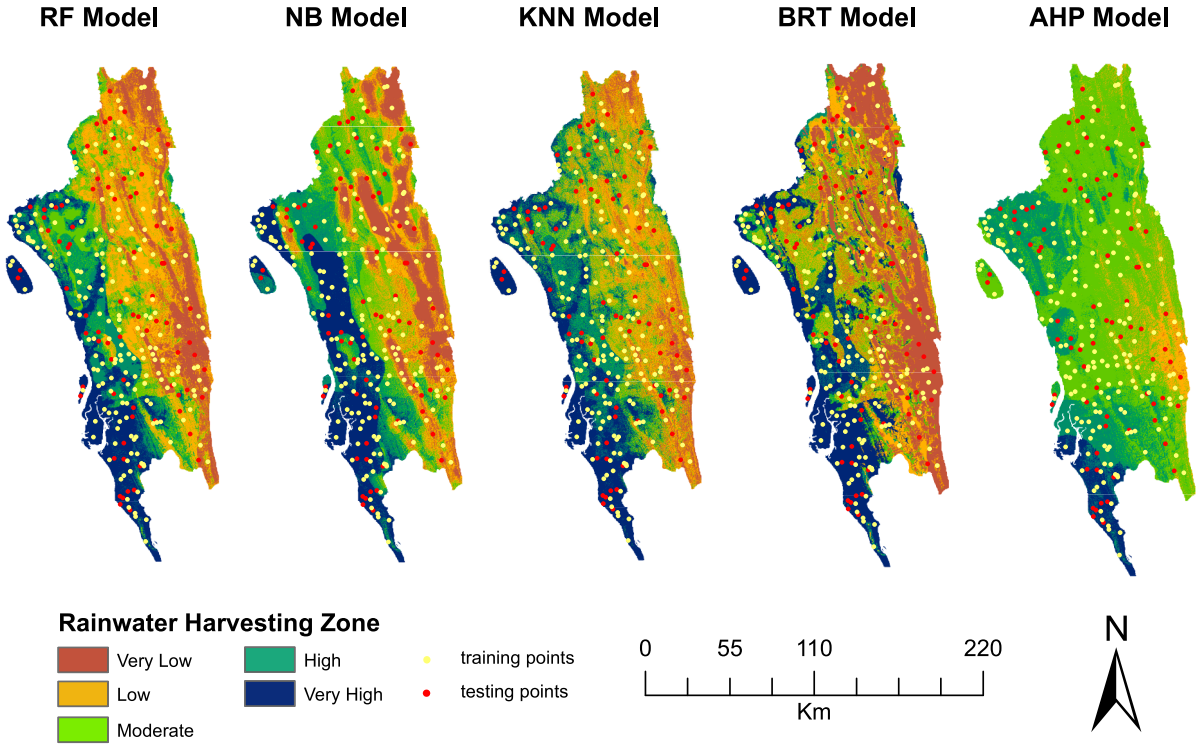


Fig. 5. The maps showing the rainwater harvesting zones using different models (RF, NB, KNN, BRT, and AHP). The harvesting zones were classified into very low, low, moderate, high, and very high potential zones. The yellow and red points in all maps represent the training and testing points for machine learning model prediction.

Table 3
Area (km^2) & Percentage (%) of each Rainwater harvesting class for the RF, KNN, NB, BRT and AHP model.

	BRT		KNN		NB		RF		AHP	
Classes	**Area**	**Percentage**	**Area**	**Percentage**	**Area**	**Percentage**	**Area**	**Percentage**	**Area**	**Percentage**
Very Low	6091.45	31.03	2051.84	10.45	3659.31	18.64	3708.85	18.89	27.69	0.14
Low	4096.69	20.87	5767.85	29.38	3958.80	20.17	5927.97	30.20	1793.66	9.14
Moderate	3149.35	16.04	3517	17.92	4446.36	22.65	3311.90	16.87	12169.54	61.99
High	1264.86	6.44	3797.88	19.35	3301.65	16.82	3150.76	16.05	4675.81	23.82
Very High	5028.47	25.62	4496.27	22.90	4264.71	21.72	3531.35	17.99	964.11	4.91

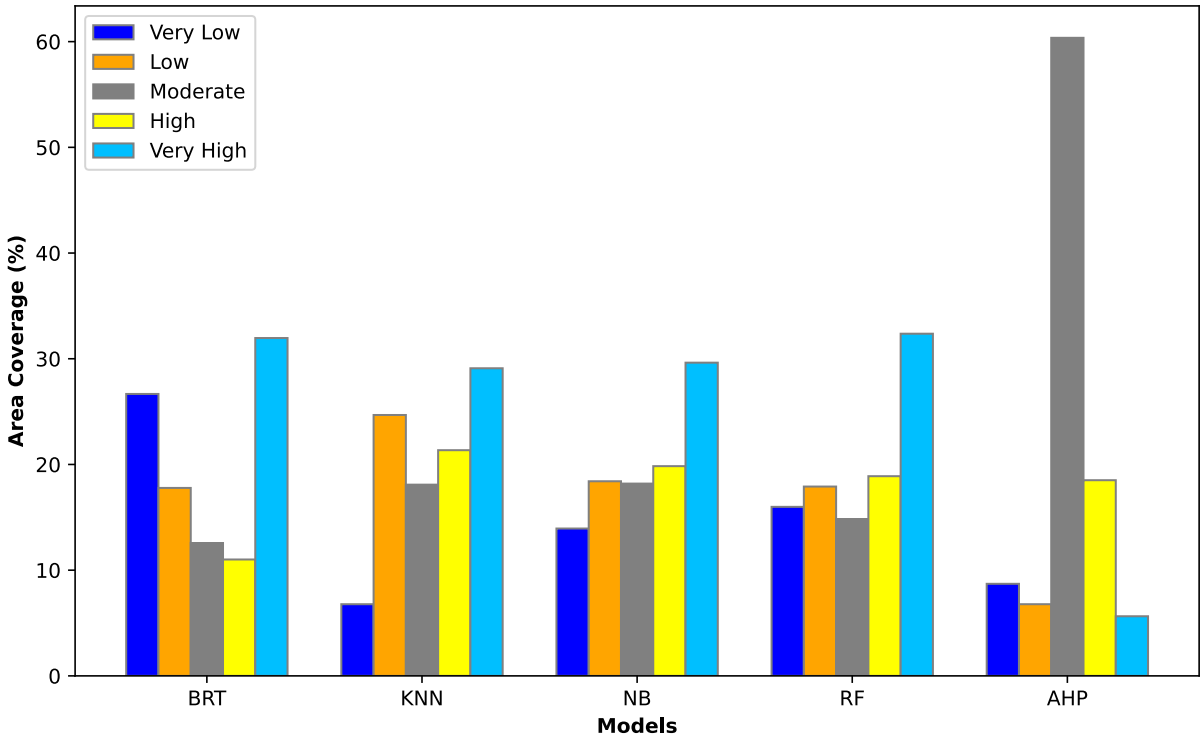


Fig. 6. The bar diagram shows the area coverage percentage of very high, low, moderate, high, and very high potential RWH zones. The five models (BRT, KNN, NB, RF, and AHP) used for RWH zones suggested the percentage of area for RWH potential zones. All models classified the study area into five RWH potential zones using eleven parameters. The different colours show the different potential zones in the study area.

3.4. Variable importance factor

Various conditioning factors influence the predictive zone maps for RWH. This study employed eleven conditioning factors to generate these maps using four ML algorithms (e.g., RF, KNN, NB, BRT) and a geospatial approach (e.g., AHP). The variable influence factors were calculated based on the two best-performing ML models (Fig. 8).

The analysis of variable importance for the BRT and RF models underscores distinct approaches to feature significance in predicting RWH potential in the Chattogram hilly districts. The BRT model shows a pronounced reliance on two primary features, drainage density and elevation, which account for 22.81 % and 22.04 % of the model's importance, respectively. This suggests that the BRT model's predictive accuracy heavily depends on these key variables, indicating a more focused yet potentially less robust model if these variables were inaccurate or unavailable.

Conversely, the RF model displays a more distributed allocation of importance across multiple features. The highest contributing feature is rainfall, which accounts for 15.73 % of the importance, followed closely by other significant features such as drainage density, elevation, aspect, TWI and slope, which account for 15.55 %, 11.45 %, 9.35 %, 8.82 %, and 8.34 %, respectively. This distribution reflects a more holistic incorporation of diverse features, potentially enhancing the model's robustness and generalizability.

Thus, while the BRT model is driven by a few dominant factors, the RF model's predictive power is derived from a more balanced and comprehensive consideration of multiple features. This difference highlights the varying methodological strengths of BRT and RF in the context of environmental modeling and resource management.

4. Discussion

This study successfully employed a novel approach by integrating ML algorithms and geospatial techniques to evaluate RWH potential in the CHT, Bangladesh. The findings provide valuable insights for sustainable water management practices in this region characterized by water scarcity and a growing population.

The spatial distribution of RWH suitability across the Chattogram hills has a pronounced north–south gradient. The analysis consistently determines that the northern region is less suited, the middle region shows a range of low to moderate suitability, and the southern region exhibits the maximum potential. This pattern corresponds to the well-documented impact of geographical features on the practice of collecting rainwater. Regions with higher altitudes in the northern areas have more pronounced inclines, resulting in swifter surface water flow and reduced water penetration. In addition, the more level terrains and

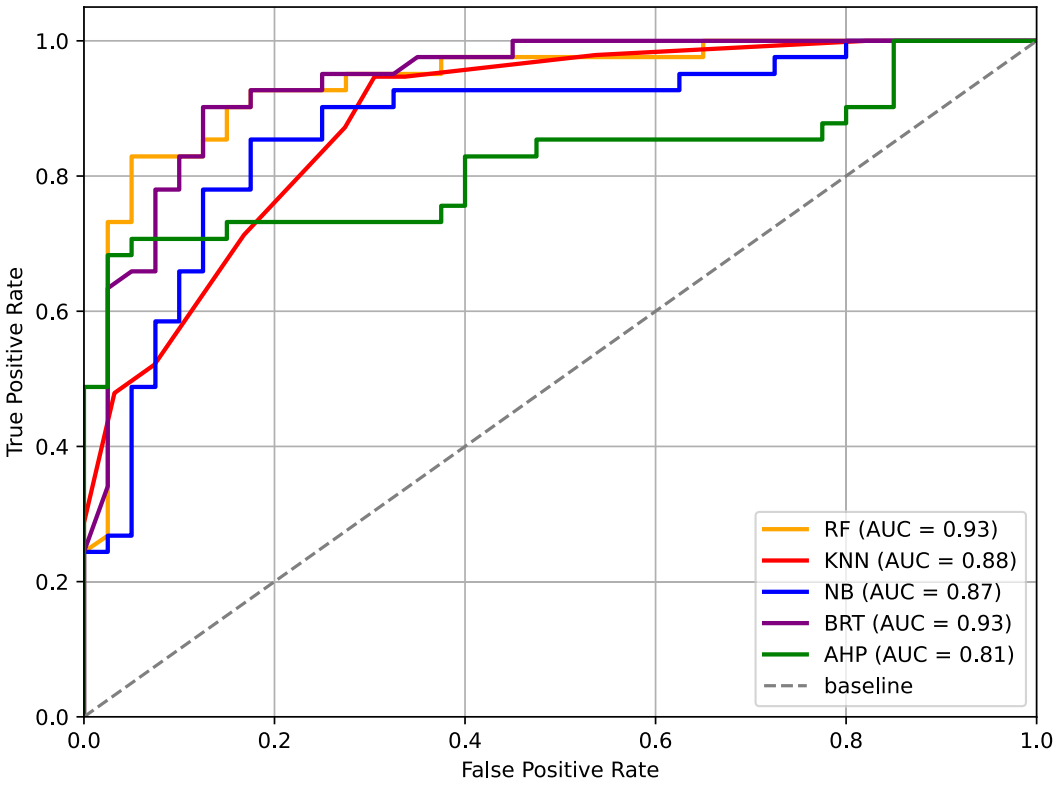


Fig. 7. The AUC represents the model's ability to discriminate between positive and negative classes. Its value ranges from 0 to 1. The values near 1 represent perfect models. The RF and BRT models show the highest AUC values of 0.93. The AHP model shows the lowest AUC values among the five models. The different colours show true positives and false negative rate of different models (RF, KNN, NB, BRT, and AHP). The dashed line shows the baseline of AUC.

Table 4
Validation by Area under the curve (AUC) of five Rainwater harvesting models.

Model	AUC
BRT	0.93
KNN	0.88
RF	0.93
NB	0.87
AHP	0.82

decreased altitudes in the southern regions facilitate improved water retention (Brimicombe, 2009). These findings align with other research conducted in hilly areas globally, emphasizing the significance of taking into account variations in topography while assessing the potential for RWH (Asif et al., 2024; Jha et al., 2014; Lani et al., 2018; Misra et al., 2024).

The research highlights the importance of geographical directionality in determining the possibility of rainwater gathering. Although all models showed a trend from north to south, the specific zones of applicability differed among the models (Table 3). This highlights the impact of the selected algorithms and the assumptions they are based on. The BRT model, in contrast to the RF model, accurately projected a far more prominent "very high" suitability zone in the southern region. This discrepancy can be ascribed to the BRT's emphasis on robust feature interactions, which may enhance the impact of particular characteristics such as drainage density or lineament density in specific geographical regions (Hulse and Khoshgoftaar, 2009; Soultan and Safi, 2017). These data indicate that choosing the appropriate ML models might affect the identification of the best rainwater-collecting zones. This emphasizes the need to consider the models' properties carefully during the study.

The examination of the five models included in the study yielded valuable insights into their effectiveness in mapping the viability of rainwater collection. BRT and RF models showed the best AUC scores. This reveals their knack for telling apart different suitability groups. Other studies back this up, proving that these methods work well for environmental modelling (Ahmed et al., 2019; Gaurihar et al., 2023; Huang et al., 2015). On the flip side, AHP had the lowest AUC. Du et al. (2024) also suggested that the ML algorithm (e.g., weight of evidence (WOE)) performs better than the AHP model. This hints at its trouble grasping the tricky links between factors and RWH potential. AHP's focus on comparing pairs might lack the mark when it comes to the web of ties among lots of environmental variables (Leake, 2000; Liaw and Chiang, 2014). It just cannot seem to get its head around all those moving parts.

The examination of variable importance factors further distinguished the methodologies of the BRT and RF models. The RF model was performed over the ANN model suggested by Halder and Bose (2024). Kumar et al. (2021) also suggested RF over the AHP for groundwater potential zoning. The BRT model mainly depended on a small number of prominent characteristics, such as elevation and slope. This means that if these most crucial variables were incorrect or missing, the model would not only become error-prone but also lose reliability. On its part, the RF model spreads importance across more variables, such as geology, drainage density, and LULC. As explained, in directly considering more factors, its capability to extrapolate to more instances in a real situation and its robustness against uncertainties might be enhanced (Nhu et al., 2020; Olsen et al., 2015). The implications of such findings are about understanding the basic choice processes of ML algorithms in a bid to ensure maps of suitability that will rend solid upon which to disseminate harvesting rainwater.

The research provides better insights into RWH suitability but needs to acknowledge its limitations. These limitations guide future research and help us understand the results. Firstly, the accuracy of RWH suitability maps depends on the data. Potential limitations in data

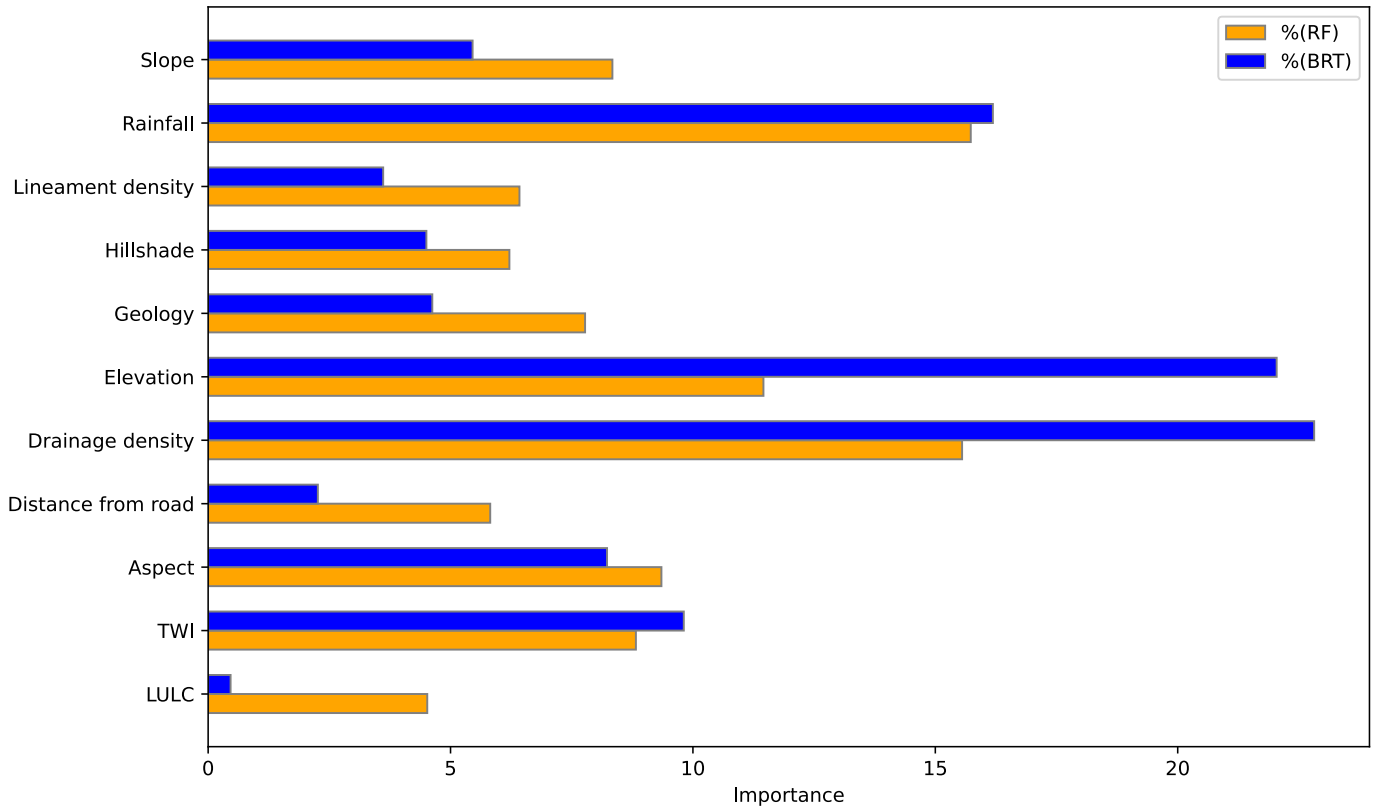


Fig. 8. The bar diagram shows the contribution of parameters in RWH zone identifications, specifically the contribution of parameters for RF and BRT, as these two models have the highest AUC values. All eleven parameters (Slope, Rainfall, Linement density, Hillshade, Geology, Elevation, Drainage density, Distance from road, Aspect, TWI, and LULC) represent the contribution to the RWH potential zone identifications. The Rainfall, Elevation, and drainage density have the highest contribution for both RF and BRT models.

resolution, accuracy, or temporal coverage can introduce uncertainty to the model outputs (Her and Seong, 2018). Secondly, while BRT and RF models performed well, their 'black box' nature makes it difficult to understand the relationships between the factors and RWH potential. Researchers can explore ways to improve model interpretability and get more insight into the decision-making process (Lundberg and Lee, 2017; Marcinkevičs and Vogt, 2023). Thirdly, socio-economic factors like land ownership, water use practices, and community acceptance play a big role in successful RWH system implementation. Lastly, given climate change, we need to assess the long-term sustainability of RWH systems under changed rainfall patterns and increased extreme weather events (Field et al., 2012; IPCC, 2012).

Further, other advanced ML techniques, such as deep learning (DL) algorithms, can be applied to improve the accuracy and understand the plausible complex relationships influencing RWH (Jain et al., 2020).

5. Conclusions

The appropriate ML algorithms were identified in this study to search the RWH zone. The BRT and RF performed better than those of other ML algorithms used in this study. Due to climate change and alterations in regional climates, many areas around the world are experiencing a decline in access to freshwater resources, both at the surface and groundwater levels. In Bangladesh, the hilly regions encompassing the districts of Rangamati, Khagrachari, Bandarban, and Chattogram are particularly affected. The elevated terrain of these regions hinders access to groundwater, while the undulating hilly landscape makes it challenging to access surface water. This study utilized a GIS-based ML algorithm alongside an MCDM approach to identify optimal zones for harvesting rainwater, aiming to alleviate pressure on surface and groundwater resources. Evaluating 11 parameters, the research found that among five models, the BRT and RF demonstrated the highest accuracy, followed by KNN, NB, and AHP. The research seeks to propose a solution to water scarcity by identifying suitable locations for RWH. It is anticipated that stakeholders and policymakers will utilize the findings to take proactive measures in developing a comprehensive solution to the water issues in the study area. Despite the availability of more advanced tools, such as hybrid learning models and DL models, this research employed ML and MCDM models, which represent a limitation of the research. The results demonstrate the efficiency of the proposed framework, as evidenced by its superior accuracy relative to other contemporary techniques. The results demonstrate the usefulness of the proposed framework, evidenced by its superior accuracy relative to other contemporary techniques. Remedial strategies for alleviating water scarcity by rainwater gathering are examined for mountainous areas. This research paper adopts a holistic strategy to tackle water-related issues, providing a replicable framework relevant to the region, particularly emphasizing other topographical areas. This framework will be further developed by investigating the integration of ML methods using ensembles with a broader range of datasets. Moreover, using the appropriate algorithm, the best place for harvesting rainwater could be chosen in the present study area.

CRediT authorship contribution statement

Md. Mahmudul Hasan: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Validation, Visualization, Roles/Writing - original draft, Writing - review and editing. **Md. Talha:** Data curation, Methodology, Formal analysis, Validation, Roles/Writing - original draft. **Most. Mitu Akter:** Conceptualization, Visualization, Data curation, Writing - review and editing. **Md Tasim Ferdous:** Methodology, Data curation, Formal

analysis, Validation, Visualization, Roles/Writing - original draft, Writing - review and editing. **Pratik Mojumder:** Data curation, Visualization, Roles/Writing - original draft, Writing - review and editing. **Sujit Kumar Roy:** Conceptualization, Methodology, Project administration, Resources, Writing - review and editing. **N. M. Refat Nasher:** Conceptualization, Supervision.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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